

OVHcloud's carbon calculator methodology

Digital technology is playing an increasingly important role at the heart of our professional and personal lives. The democratization of cloud services accompanying this explosion in usage is contributing more and more to the digitization of all economic sectors, creating a cycle of continuous improvement and change.

For over 20 years, OVHcloud has been innovating to make the cloud more sustainable by reducing the impact of its datacenters and servers and using resources sparingly.

Maintaining this innovation momentum, OVHcloud is now offering a carbon calculator designed to provide an estimation of greenhouse gas emissions. It stands out as the first to offer a transparent and exhaustive approach. Allowing OVHcloud customers to understand their carbon footprint, the results linked to their cloud usage are broken down into 3 emission categories:

The manufacturing phase: upstream greenhouse gas emissions linked to the purchase and assembly of components of the server(s);

The use phase: emissions linked to electricity consumption of the server(s);

Other emissions: other emissions such as freight or the impact induced by employees - known as «operations».

IJO, a consultancy specializing in green IT, and Cost House, a consultancy specializing in economic performance, have been commissioned to carry out a methodological review of the OVHcloud carbon footprint simulator using the Bilan Carbone® method. You will find [here](#) the certification letter for the carbon footprint calculator, certified as compliant with the Bilan Carbone® method, meeting the requirements of the GHG Protocol and, moreover, adapted to OVHcloud's activities, presenting a high level of accuracy.

Before displaying this calculator, and in order to carry out this project successfully, a strict methodology has been established. We present here a short version of it.

I. Methodology

1. OVHcloud's methodological choices

a. Commitment in providing transparent, comprehensive, and reliable information

OVHcloud's commitment in communicating carbon-related information to its customers begins with the calculation of the GHG (Greenhouse Gas) emissions associated with all the delivered services. OVHcloud develops an approach that embraces all products of its catalog in order to provide a complete carbon footprint to each client. Every service delivered has its own features and addresses specific customer needs. Therefore, developing calculations is a process that OVHcloud handles thoroughly in a methodical and structured way, embraced in overall sustainability roadmap.

First of all, as recommended by the highest standards in GHG accounting, OVHcloud also includes in this tool, emissions related to OVHcloud operations. The aim here is to provide trustworthy results and highlight commitment to reducing all of OVHcloud emissions. Details about actions on this matter can be found in OVHcloud yearly carbon assessment.

The choice made by OVHcloud is to communicate to the customers all of the emissions from the three scopes of carbon reporting (scope 1 – direct emissions, scope 2 and 3 – indirect emissions). Transparency towards customers is key, and therefore it is OVHcloud responsibility to provide a complete carbon footprint by categories to give to the clients all the elements to monitor and reduce the impacts of the products they use.

OVHcloud activities' main sources of GHG emissions are the production of servers (22 % in 2023) and the supply of electricity (45 % in 2023) – mainly for the data centers. The rest of the scopes of emissions account in total for 33 %. To enable customers to actually reduce their emissions, and in accordance with OVHcloud transparency values, the carbon calculator model shows a carbon footprint allocated over three pillars: Manufacturing / Electricity / Operations (Figure 1).

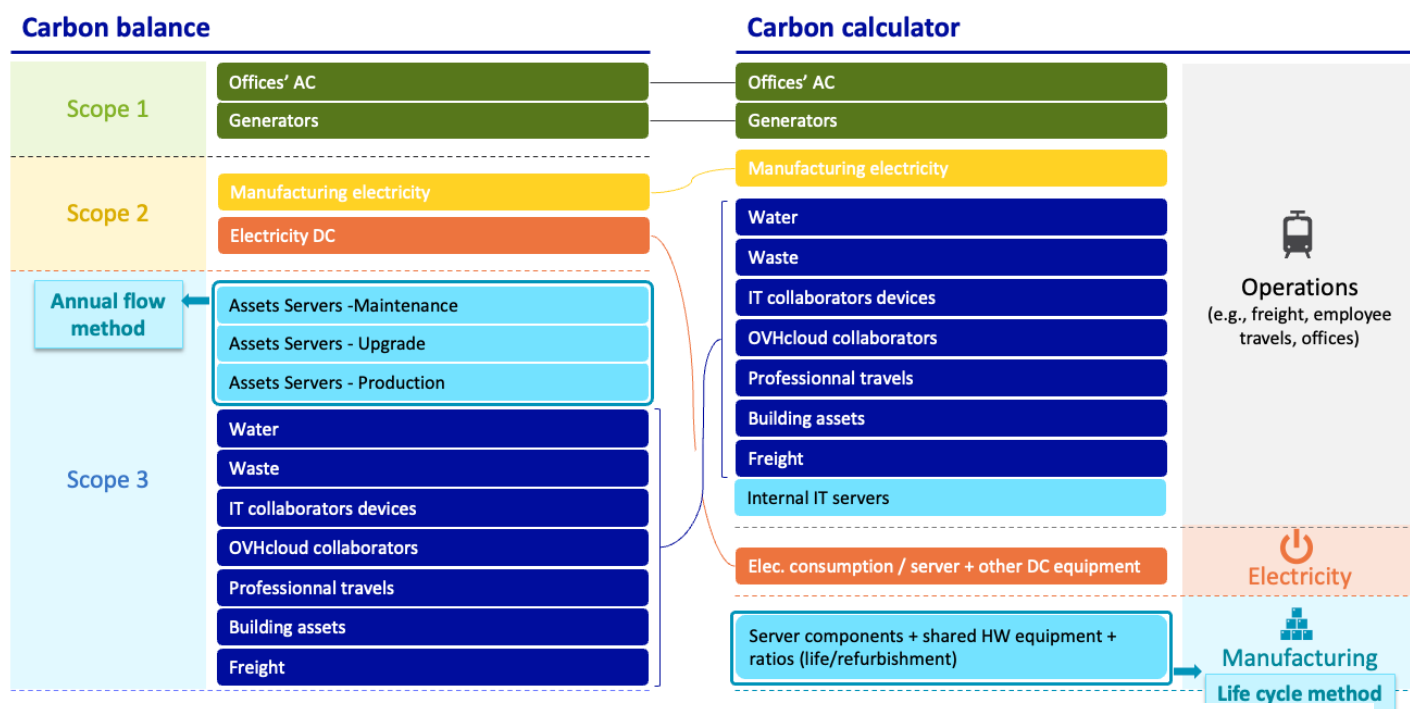


Figure 1. Perimeters for assessing the GHG carbon emissions of Cloud services (Source: OVHcloud)

Key insights

3 pillars have been chosen to show the environmental impacts:
Manufacturing, Electricity, Operations

b. Development process and selection of products addressed

OVHcloud offers a large range of services from computing to data storage or web hosting. This extensive catalog is a challenge when addressing the allocation of GHG emissions. Consequently, and with the aim of designing a coherent and reliable approach, we decided to address five major products:

- Baremetal dedicated servers (first available version: summer 2023)
- Hosted Private Cloud (first available version: November 2023)
- Public Cloud – Compute
- Public Cloud – Object Storage
- Web Hosting

As published in the Fiscal Year 2021 Annual Results¹, the revenues of these products account for about 74 % of the global revenues.

The environmental display tool will include the precise calculation of the emissions related to these products in all of locations. For the first version of the tool, a complete methodology for Baremetal dedicated servers has been developed as presented below.

¹ Fiscal Year 2021 Annual Results are available on OVHcloud corporate website: <https://corporate.ovhcloud.com/en/investor-relations/financial-results-2021/>

2. General approach for calculating customer carbon footprint

Developing a model to assess the environmental impacts of products and manage their allocation to each customer requires being able to evaluate the carbon footprint of each Baremetal server installed in OVHcloud's data centers that is used for the operation of customer services. In the calculation model developed by OVHcloud and presented in this paper, the estimation of the servers-related carbon emissions is a key enabler for the rest of the process, including spreading the emissions over the different services used by the customers, in accordance with the configuration and the location of the servers.

GHG emission calculation is made at the server level and includes all the materials, teams and activities required for operating the building and managing the servers. Furthermore, as it is explained in the sections below, the methodology differentiates two different types of products: dedicated services and mutualized ones (more precisions are given in the sections "Methods for allocating dedicated server's carbon footprint"). The global approach explained and specified in this paper is summarized in Figure 2.

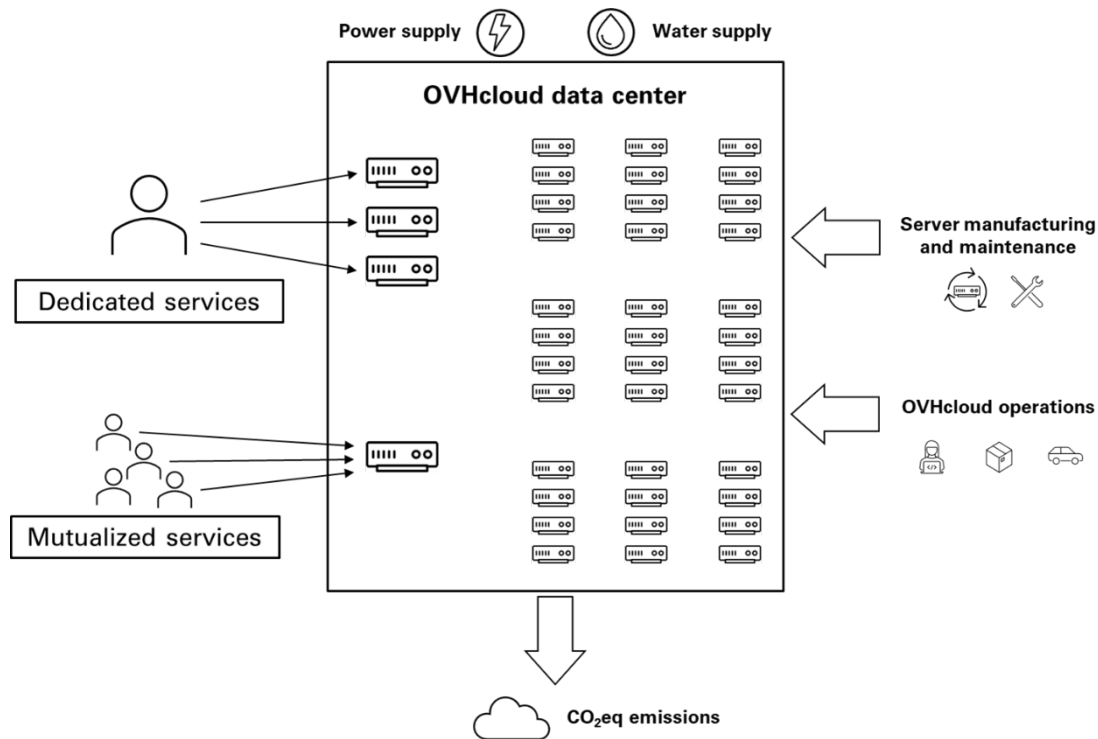


Figure 2. Diagram of the global approach chosen for carbon footprint assessment

The three pillars (Manufacturing, Electricity and Operations) of OVHcloud carbon calculator model also reflect three different approaches to calculate carbon footprints. One is due to on-site activities of manufacturing which require various materials; the second is directly linked to the location of OVHcloud's buildings, the available energy sources and OVHcloud's efforts towards the use of low-carbon energy; and the last one gathers a large range of emission sources to allocate to each customer. This section aims at providing key elements on the three approaches developed by OVHcloud to assess customer related carbon footprint.

3. Approach used for the calculation of the carbon footprint by server according to three natures of emissions

a. Manufacturing

As it represents 22 % of the global carbon assessment of the company, the manufacturing of the servers (including upstream emissions of the hardware manufacturing) used in OVHcloud data centers is central in the calculation of the carbon footprint for the services provided to customers. OVHcloud assembles all of the servers we use and their cooling system, with various configurations. The wide range of configurations, designed for the different use cases of the customers, requires a strong methodology to be able to evaluate the carbon footprint of every server used in each data center.

OVHcloud has indeed implemented high standards to assemble and disassemble servers. Therefore, components have different use phases. This integrated model enables OVHcloud to provide price competitive services with a high level of reliability while minimizing the associated environmental impacts.

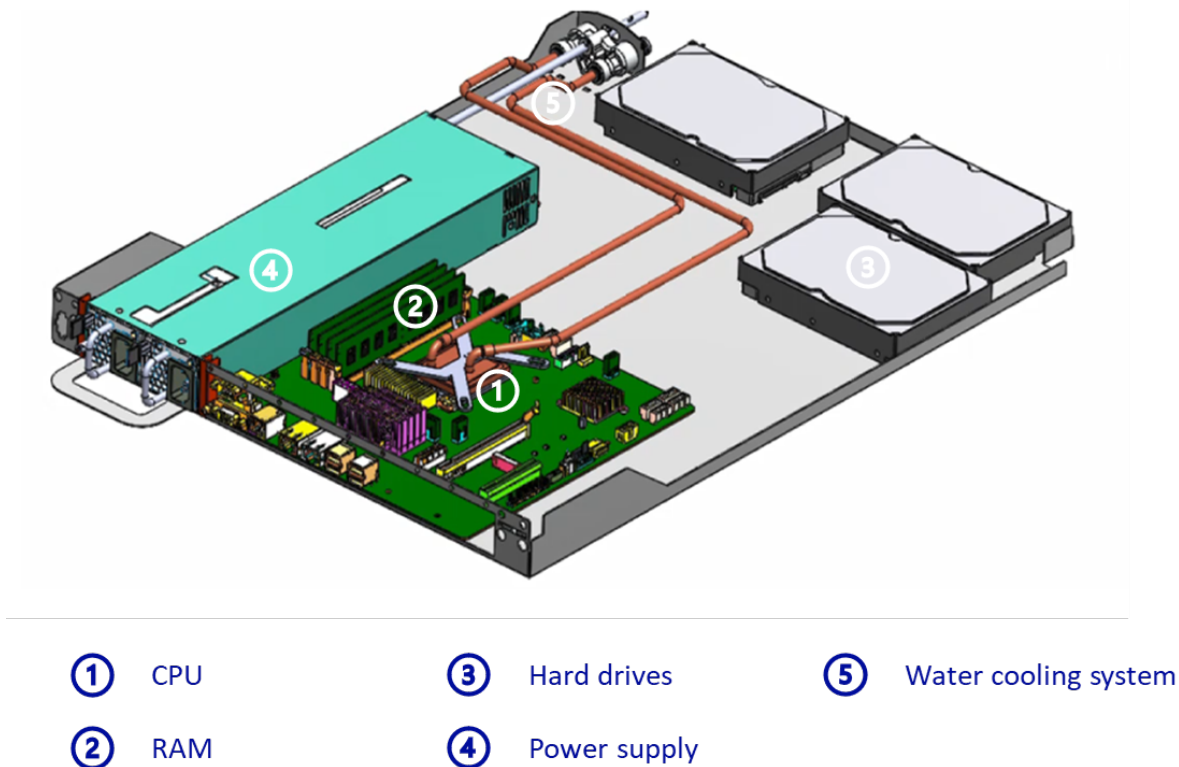


Figure 3. Explaining picture of the composition of an OVHcloud server

The carbon impact of producing a server predominantly comes from the manufacturing of a few main types of components such as CPU, GPU, RAM, HDD/SDD, motherboard and power supply (Figure 3). Therefore, it was important to develop a model that reflect the real impacts and therefore that is based on the physical configuration of the machine. As OVHcloud assemble every server, the model has to be fed by reliable and regularly updated data.

Table 1. GHG footprint of component (Source: IJO)

Component type	Unit	Emission factor	Mark-up for doubt
Rack size	1U / 2U / 3U	200 / 250 / 350 kgCO ₂ eq	60 %
CPU size	Number of physical cores	1,5 kgCO ₂ eq/physical core	
RAM size	Giga byte	2 kgCO ₂ eq/Gbyte	
SSD size	Tera byte	60 kgCO ₂ eq/Tbyte	
HDD size	Tera byte	25 kgCO ₂ eq/Tbyte	

A calculation model based on the configuration of a server has been developed to assess the carbon footprint of every physical host. By combining recent data from the scientific literature, with information communicated by IT component manufacturers of which some of suppliers, the model succeeds to spread OVHcloud's impact over all the servers including primary manufacturing (Table 1). Hence, a single but evolving model enables providing precise personalized carbon footprint for every physical machine.

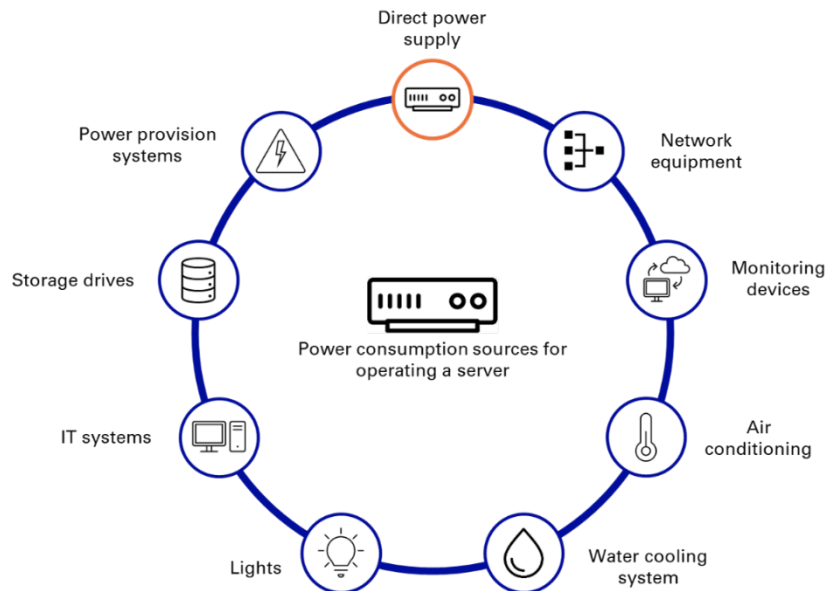
Key insights

The model developed to assess GHG emissions associated to the production of every server relies on two key points:

- The configuration of the machines, especially the CPU, RAM and hard disks, whether HDD or SDD,
- The expected lifespan.

b. Electricity

Operating a physical server requires the mobilization of many other IT materials and infrastructure which must be considered in the evaluation on the environmental impacts. A diagram summarizing the main sources of power consumption considered for estimating the direct and indirect consumptions of a server is provided in Figure 4.

**Figure 4. Main power consumptions sources for operating a server**

OVHcloud achieves to maintain low carbon emissions thanks to the location of many data centers in countries where the GHG intensity of the electricity mix is relatively low such as France or Canada. This leads to globally low CUE (Carbon

Usage Effectiveness). Also, OVHcloud achieves good results in PUE (Power Usage Effectiveness) and WUE (Water Usage Effectiveness) with the implementation of its own cooling technologies (Table 2). When we are not able to measure the PUE, we use the average PUE FY2023 of 1.29.

Table 2. GHG intensity of national electricity mix (October 2023)

Datacenter		Source of Location-based Electrical Emission Factor	Market-based – Electricity emissions (kg CO ₂ /kWh)	PUE	WUE (l/kWh)
Beauharnois	BHS	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Québec - 2022	0,0250	1,24	1,00
Erith	ERI	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Great Britain - 2022	Refer to location based electrical emission factor	1,25	0,20
Gravelines	GRA	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - France - 2022	0,0356	1,22	0,13
Hauts de France	HDF	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - France - 2022	Refer to location based electrical emission factor	-	-
Hillsboro	HIL	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Portland General Electric Company - 2022	Refer to location based electrical emission factor	1,46	0,93
Limburg	LIM	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Germany - 2022	0,0250	1,31	0,27
Mumbai	MUM	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Western India - 2022	Refer to location based electrical emission factor	-	-
Roubaix	RBX	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - France - 2022	0,0356	1,35	0,30
Singapore	SGP	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Singapore - 2022	Refer to location based electrical emission factor	-	-
Strasbourg	SBG	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - France - 2022	0,0356	1,25	0,57
Sydney	SYD	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - New South Wales - 2022	Refer to location based electrical emission factor	-	-
Vint Hill	VIN	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - PJM Interconnection, LLC - 2022	Refer to location based electrical emission factor	1,37	0,04
Warsaw	WAW	Electricity Maps - Data Portal. (2023, July 1). https://www.electricitymaps.com/data-portal - Poland - 2022	0,0126	1,32	0,36

* Made available here under the Open Database License (ODbL) : <https://opendatacommons.org/licenses/odbl/1-0/>

The power consumption is not equal between each OVHcloud server. It depends on its configuration and on its usage. OVHcloud is currently deploying Smart PDUs to monitor precisely the electricity consumption of each server.

As all servers do not benefit from this monitoring, a simplified approach has been used in the calculation model. For each server, OVHcloud has evaluated the associated power consumption with an internal project aiming at providing for every potential configuration – capacity of the CPU, size of the RAM, etc. – and customer profile the typical power

consumption expected. The evaluated typical power consumptions are then compared to real measured power consumptions to validate the models.

Nevertheless, as explained above assessing the net power consumption of the server has to be completed by the rest of the electricity consumption of the datacenter. To provide the best quality and most useful information to customers, we consider the proper PUE of each datacenter. This is done considering the overall power consumption of the data center and distributing extra power consumptions – those that are not from direct server consumptions – over all server connected in the building. The method allows for a more precise estimation of the power consumption to be allocated to each OVHcloud server.

$$PUE = \frac{\text{Energy consumption of Datacenter}}{\text{Energy consumption of IT equipment}}$$

Equation 1. PUE2 rolling 12-month average with ISO30134 norm

OVHcloud leads an ambitious green energy policy. Once the power consumption has been estimated, the GHG emissions are calculated on a location-based approach, meaning that the local energy mix is taken into account for each datacenter (Table 2). Since November 2023, we have been taking into account the GHG emissions corresponding to our low carbon energy contracts as indicated by the "market-based" emission factors. At this stage, the carbon footprint of the usage phase of each server completed by other IT materials and infrastructure is assessed and included in the global results.

c. Operations

Carbon Balance

As explained above, this nature of emissions gathers every OVHcloud GHG emissions that is not accounted as either Manufacturing or Electricity. The operations-related carbon footprint is the sum of the impacts of several scopes and activities including:

- Scope1: Direct emissions from
 - Back-up Power Generators
 - Diffuse Emissions of HVAC
 - Internal fleet of combustion engine vehicles
- Score 2: Indirect emissions from
 - Electricity consumed for server manufacturing
- Scope 3: Indirect emissions from
 - Energy and fuel related activities
 - Freight and Transportation
 - Building immobilization and capital goods
 - Business travel
 - Employee commuting
 - End of life treatment
 - Backbone & Network assets

- Goods and services purchased for OVHcloud Operations (Water, Employees computers, Internal information system)

The data considered for this nature of emissions comes from the carbon balance of the company².

The method chosen is a simplified approach using an equal distribution of the Operations impacts over the entirety of the servers dedicated to customer use.

OVHcloud's internal IT

This raises the question of the allocation of the impacts generating by OVHcloud's internal IT. It has been decided that there would be one single methodology for distributing the related GHG emissions, and that this method would be the same as for the Operations. Indeed, when dealing with corporate IT equipment the impacts must be equally allocated over all OVHcloud servers.

Key insight

The environmental impact of a server dedicated to internal usage is distributed over the servers at a global view.

As for the Operations, OVHcloud Sustainability teams are currently working on a different allocation method that would more realistically reflect the server resources consumptions. However, for the time being, as the operation part represents a minor part of the total footprint, the equal distribution will give satisfactory results. As an example, repartition of the data centers buildings according to the number of servers installed inside seems relevant, while allocation of the support team's footprint can be challenged: products can have dedicated teams, according to level of support for instance.

² It follows the GHG Protocol methodology. Greenhouse Gas Protocol provides standards, guidance, tools and training for business and government to measure and manage climate-warning emissions.

II. Methods for allocating dedicated server's carbon footprint

The whole methodology developed by OVHcloud and presented here aims at distributing the entirety of the company's GHG emissions over the customer carbon invoices. It is assumed that the whole company activity is dedicated to delivering the best services to its customers, and therefore all of its impacts must be considering in this reporting process.

Once the impacts of all the servers have been estimated, it results in a list of machines with associated amount of GHG emissions divided into three natures of emissions – Manufacturing, Electricity and Operations. To be able to report to each customer the emissions due to its use of OVHcloud products, the next step is to allocate these emissions over the OVHcloud's customers considering the services they have subscribed to and the infrastructure they are based upon.

Within the products whose footprint is detailed in the calculation tool, OVHcloud offers two different types of products:

- products that imply dedicated equipment to a customer– for a server, it means that the whole physical machine capacity and therefore all virtual machines hosted are dedicated,
- products hosted on servers which are mutualized between different customers.

The categorization of OVHCloud's offers is given in Table 3.

These two kinds of offers must be considered separately in the reporting process as the carbon footprint allocation rules will vary and require different assumptions. The two approaches are presented here below.

Table 3: Table of OVHcloud offers with the categorization dedicated vs. mutualized services

OVHcloud dedicated products	OVHcloud mutualized products
Baremetal	Public Cloud – Compute
	Public Cloud – Storage
Hosted Private Cloud	Public Cloud – Network
	Web Hosting

Regarding the dedicated services, the customers have at their disposal a range of physical machines which can be operated whenever they want and to the capacity they want. It is then evident that for such services the whole carbon footprint associated with the dedicated servers must be allocated to the client who holds them. Therefore, every emission whatever the nature (manufacturing, electricity and operations) would be reported in the carbon report of the client as presented in Figure 5.

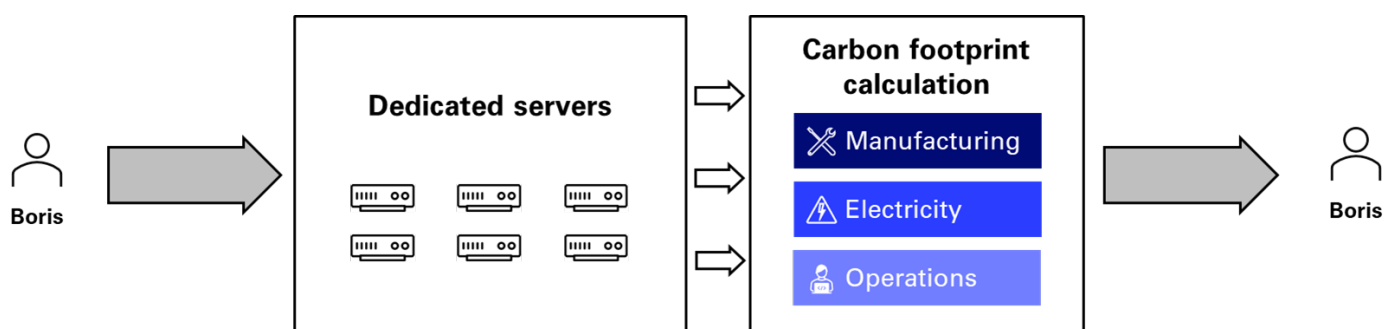


Figure 5. Diagram of the calculation approach chosen for dedicated services

Once, the list of servers dedicated to one client has been identified, it is about summing the emissions of each server related to the three categories (Operations, Manufacturing, Electricity) using the methods presented above. Finally, it is possible to present to the client a carbon footprint encompassing all 3 scopes of OVHcloud own carbon balance, with a distribution by data centers and services.

Key insight

For dedicated products (e.g., Baremetal, Private Cloud) all emissions associated with the dedicated equipment are allocated to the customer.

